THE CATHOLIC DIOCESE OF TANGA
TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)
FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1

032/2A CHEMISTRY 2A
(ACTUAL PRACTICAL)

Time: 2:30 Hours **2026**

Instructions

1. This paper consists **of two** questions. Answer **all** questions.
2. Each question carries **25** marks.
3. All writings should be in **blue** or **black** ink except diagrams, which must be in **pencil.**
4. Cellular phones and any unauthorised materials are **not** allowed in the examination room.
5. Mathematical tables and non-programmable calculators may be used.
6. Write your **examination number** on every page of your answer booklet(s).
7. For calculations you may use the following constants. Atomic masses:
   H = 1, O = 16, Na = 23, C = 12.

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1. Vinegar is a versatile, natural cleaner due to its acetic acid content, which breaks down grease, mineral deposits and odours, mostly diluted with water during its preparation. 8.4 Litres of cleaning vinegar solution containing the unknown concentration of acetic acid (CH_3COOH) have been prepared by diluting one litre of commercial household vinegar for the purpose of cleaning stained floors and benches in your school laboratories.

   Provided with the following solution

   AA: Solution sample of cleaning vinegar containing an unknown concentration of acetic acid

   BB: Solution of 0.1M sodium hydroxide

   Phenolphthalein (P.O.P.) Indicator

   Procedures:

   (i) Put solution **AA** into the burette

   (ii) Pipette $20cm^3$ or $25cm^3$ of solution **BB** into a $250cm^3$ titration flask

   (iii) Add two drops of phenolphthalein indicator

   (iv) Titrate this solution **AA** against solution **BB** until a colour change is observed.

(v) Repeat the procedure to Obtain three more readings .

(vi) Record your results in tabular form

Questions

(a) The colour change at the end point was from ……. to……..

(b) Calculate the mean titrate volume

(c) ….……cm^3 of solution **BB** required …….. cm^3 of solution **AA** for complete reaction

(d) Calculate the concentration of acetic acid present in the solution **AA** in mol/dm^3

(e) Calculate the concentration of acetic acid present in solution **AA** in g/dm^3

(f) Calculate the molarity of acetic acid present in the commercial household in vinegar that was diluted to obtain a cleaning vinegar in solution **AA**

2. A food company notices that milk spoils faster when left outside than refrigerated. As the company chemist, you are obligated to explain why the company should continue keeping the milk refrigerated to avoid company loss. To arrive at a scientific conclusion about the scenario, you decided to carry out an experiment on chemical kinetics. The following chemical substances are on your table for this investigation: Solution **AB:** 0.02M potassium permanganate ($KMnO_4$).

Solution **CD**: 0.3M of hydrated oxalic acid ($H_2C_2O_4.2H_2O$).

Theory.

The potassium permanganate-oxalic acid reaction is a redox reaction with autocatalytic kinetics. It is used to study real-world kinetic behaviours.

The redox reaction equation is:

$$2MnO_4^-{}_{(aq)} + 5C_2O_4^{2-}{}_{(aq)} + 16H^+{}_{(aq)} \longrightarrow 2Mn^{2+}{}_{(aq)} + 10CO_2{}_{(g)} + 8H_2O{}_{(l)}$$

Procedure.

(i) Make a water bath using a 500 cm^3 beaker.

(ii) Measure 10 cm^3 of both **AB** and **CD** solutions using the measuring cylinders given, then put the solutions **AB** and **CD** in the boiling test tubes.

(iii) Put the thermometer in one of the test tubes, use the test tube holders to keep the test tubes in a water bath, and heat the contents of the test tubes up to 80°C. iv. When the 80°C reading is reached, take out the two test tubes and pour the contents into a 100 cm^3 beaker and immediately start the stopwatch and observe the colour change of the reaction mixture.

(iv) Stop the stopwatch when the purple colour disappears to colourless. Record the time used.

(v) Repeat procedures ii to v three more times by heating the test tube contents to 700°C, 600°C, and 500°C, respectively, as shown in the table below.

Exp.	Volume of AB Cm^3	Volume of CD Cm^3	Temperature ^0C	Time(t) sec	1/t (sec^{-1})
1	10	10	80		
2	10	10	70		
3	10	10	60		
4	10	10	50		

Questions:

(a)

(i) Complete the table by filling the blank columns.

(ii) What was the aim of this experiment?

(iii) What is a redox reaction?

(iv) With a reason, state which substance was reduced in the experiment between potassium permanganate ($KMnO_4$) and oxalic acid ($H_2C_2O_4$).

(v) Name the property that you used to measure the rate of the chemical reaction of this experiment.

(b)

(i) What does 1/t represent?

(ii) What was the factor affecting the rate of this experiment?

(iii) Plot a graph of 1/t against temperature.

(c) Basing on your graph you have drawn in (b) (iii), what conclusion can you present to the company CEO regarding the continuous refrigeration of the milk products?

(d) Describe, using a chemical equation, how you can chemically test the gas produced in this experiment.